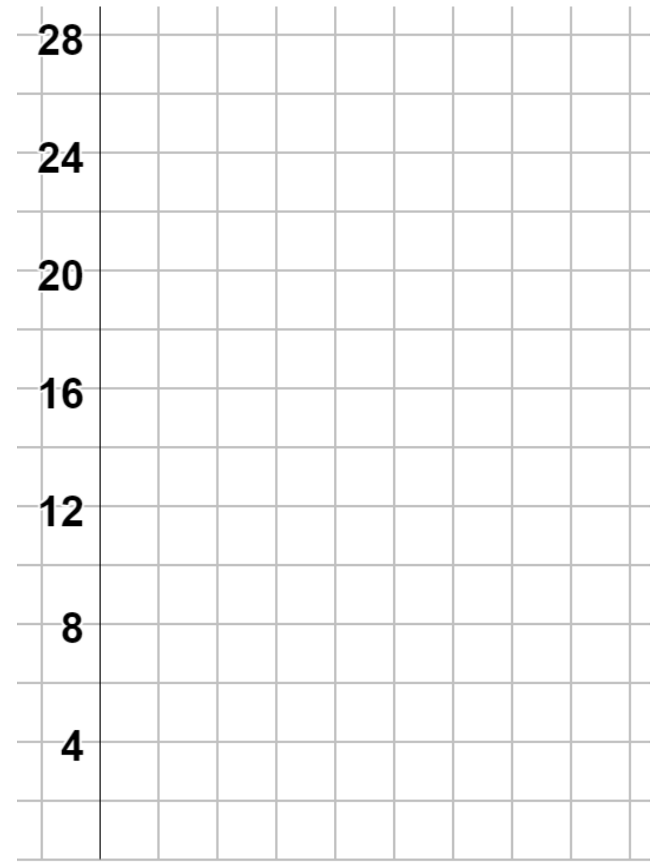
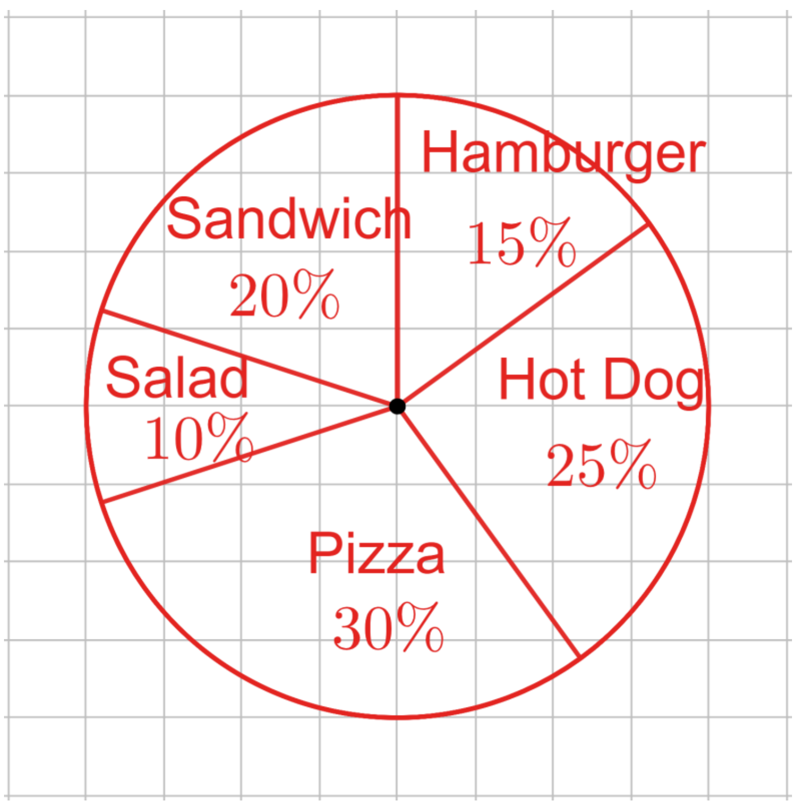
Grade 7 Unit 13 Assessment Guide

General Information	In Unit 13, students work with a variety of data sets, represented numerically and graphically, to interpret and summarize information connected to that data. Students use their understanding of measures of center and spread to select most appropriate measures and to compare two data sets. The concepts on this assessment utilize knowledge students gained in Grade 6 coursework (Units 9 and 10), so it may be helpful for the teacher to note any patterns in student successes or struggles on this assessment to determine what differentiated supports students may need. Students are allowed the use of a calculator for this assessment. Most of the questions on this assessment are not calculator dependent. Teachers should note that an assessment limit in the draft item specifications states that items that ask students to determine the mean of a data set are limited to 10 data points, and categorical data sets are limited to six categories. This restriction is likely in place so that students do not spend an extraordinary amount of time on calculations but instead focus on making sense of the measures of center and variability in the context of the problems. Question 4a has the student construct a circle graph. This question allows a teacher to monitor student progress from unit 7. Teachers who would prefer that students do a bar graph can use the grid shown here to replace the circle. If that change is made, then Question 4b should also be changed so that circle graph replaces bar graph. 
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.7.DP.1.1
a. The measure of center that should be used to represent the typical sale price of the homes in the neighborhood is the [○ mean / ○ median] home price. The typical price for the neighborhood is $179,000.	**Recommended Scoring** **(6 Total Points)**		Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students who correctly choose median but whose work has a minor error 4 points. Consider not giving students any credit for guessing median without showing any work. Note that work could be shown on the stem-and-leaf plot if the student recognizes median should be used based on the outlier.
Question	**1b**	**Benchmark(s)**	MA.7.DP.1.1
b. The data [○ has an outlier, / ○ is generally symmetric,] so [○ the range is / ○ the interquartile range is / ○ both the range and the interquartile range are] the preferred measure(s) of variation.	**Recommended Scoring** **(6 Total Points)**		Consider giving 3 points for each correct answer choice.

Question	2a	Benchmark(s)	MA.7.DP.1.2
a. The southern states have a median of 48.76 strikes/km^2 and a mean of 53.19 strikes/km^2.		**Recommended Scoring** **(10 Total Points)**	Consider giving 5 points for each correct measure of center. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**2b**	**Benchmark(s)**	MA.7.DP.1.2
b. The western states have a median of 5.46 strikes/km^2 and a mean of 6.29 strikes/km^2.		**Recommended Scoring** **(10 Total Points)**	Consider giving 5 points for each correct measure of center. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**2c**	**Benchmark(s)**	MA.7.DP.1.2
c. Neither data set has an outlier. Therefore, the [○ mean / ○ median] should represent the center and the [○ range / ○ interquartile range] should be used to represent variability.		**Recommended Scoring** **(6 Total Points)**	Consider giving 3 points for each correct answer choice.

Question	2d	Benchmark(s)	MA.7.DP.1.2
d. The western states tend to have [○ less ○ more] variability than the southern states because the [○ range ○ interquartile range ○ range and the interquartile range] is/are [○ larger ○ smaller] than the [○ range ○ interquartile range ○ range and the interquartile range] for the southern states.		**Recommended Scoring** **(6 Total Points)**	Consider giving 3 points for correctly choosing "less" in the first answer box. Consider giving 3 points for correctly choosing all three of the rest of the answer boxes. These boxes are interconnected.
Question	**3a**	**Benchmark(s)**	MA.7.DP.1.2
a. A car's stopping distance on dry pavement is typically [○ less ○ more] than the stopping distance on wet pavement. The stopping distance on dry pavement has [○ less ○ more] variability than the stopping distance on wet pavement.		**Recommended Scoring** **(6 Total Points)**	Consider giving 3 points for each correct answer choice.

Question	3b	Benchmark(s)	MA.7.DP.1.2
b. The stopping distance of car *D* on wet pavement is ○ a gap ○ a cluster ○ an outlier which means that the ○ mean ○ median is the best measure of center and the ○ range ○ interquartile range is the best measure of spread for this data.		**Recommended Scoring** **(9 Total Points)**	Consider giving 3 points for each correct answer choice.
Question	**4a**	**Benchmark(s)**	MA.7.DP.1.5
		**Recommended Scoring** **(30 Total Points)**	Consider giving 5 points for the work that leads to each percentage or measure of each sector of each circle. Consider giving 5 points for a correct circle graph. The circle graph is given on grid paper.
		Additional Notes	The circle graph is given on grid paper to help guide students. Students may notice that by grouping a few categories they will have 50%, or half of the circle. This may help students use the grid. Teachers should consider the tools provided when grading the circle graph.

Question	4b	Benchmark(s)	MA.7.DP.1.5
b. Select all of other graphical representations that would be appropriate for the data. ☐ Box plot ☐ Bar graph ☐ Frequency table ☐ Histogram ☐ Stem-and-Leaf plot		**Recommended Scoring** **(8 Total Points)**	Consider giving 4 points for each correct answer choice. Consider deducting 4 points if an incorrect answer choice is included. Consider giving no points if a student chooses all of the answer choices.
Question	**5**	**Benchmark(s)**	MA.7.DP.1.2
The data displays show that the distribution of the amount spent, in dollars, in December is [○ less than / ○ more than / ○ approximately the same as] the amount spent, in dollars, in January.		**Recommended Scoring** **(3 Total Points)**	No partial credit suggested.